[https://averroes.ai/blog/common-manufacturing-defect-types]

Name: Fortune Mondlane

Topic: Reducing Manufacturing Defects Using Data Analytics: Trend Analysis, Severity Assessment, and Cost Drivers in 2024 Quality Records

1. Introduction

1.1. Background

Manufacturing defects are faults, imperfections, or inconsistencies in products that arise from errors during the production process (Owen, 2002). Although defects may not be completely eliminated due to inherent process variability, it is crucial to minimise them to protect safety, maintain product quality, and improve organisational profitability. Defects can lead to serious consequences such as rework, scrap, production delays, customer complaints, and product recalls. For example, the American automotive industry recalled approximately 30 million vehicles in 2004 due to defective components and quality failures (Lucky & Takim, 2015). In addition, manufacturers may face legal liability for injuries caused by defective products, further increasing the cost of poor quality (Owen, 2002).

To address these challenges, organisations adopt structured quality improvement methodologies such as Six Sigma, which aims to reduce process variation and defects through statistical and systematic problem-solving approaches. A widely applied Six Sigma framework is DMAIC (Define, Measure, Analyse, Improve, and Control), which supports root cause identification and long-term defect prevention (Sadikin, 2023). With the increasing availability of quality inspection data, defect reduction strategies can be strengthened through data-driven analysis of defect trends, severity levels, defect locations, and repair costs.

1.2. Motivation

Manufacturing defects contribute significantly to the cost of poor quality through rework, scrap, and repair costs. Despite many organisations collecting defect inspection records, these datasets are often underutilised, and patterns such as high-cost defect types, recurring defect locations, and inspection effectiveness remain unidentified. This creates an opportunity for data-driven defect reduction, where exploratory data analysis and statistical evaluation can be applied to support better decision-making, prioritise improvement actions, and reduce defect-related losses.

1.3. Aim and Objectives

This project aims to analyse a 2024 manufacturing defects dataset using statistical and data analytics techniques in order to identify defect trends, major defect cost drivers, and potential defect reduction strategies.

Objectives:

- To perform data cleaning and preprocessing to ensure the dataset is suitable for analysis.
- To conduct exploratory data analysis (EDA) to understand the distribution and relationships between defect variables.

- To identify the most frequent defect types and defect locations using descriptive statistics and Pareto analysis.
- To evaluate the relationship between defect severity and repair costs.
- To compare inspection methods and determine their effectiveness in detecting severe and high-cost defects.
- To develop data visualisations that clearly communicate defect trends and quality improvement priorities.

1.4. Scope and Limitations

This project focuses on the analysis of a manufacturing defects dataset obtained from Kaggle. The dataset contains defect records for the year 2024, including defect date, inspection method, defect type, defect location, severity level, and repair cost.

Limitations:

- The dataset is cross-industry and not specific to a particular manufacturing sector, which may limit the ability to draw industry-specific conclusions.
- The dataset only covers the year 2024, therefore long-term defect trends across multiple years cannot be evaluated.
- The dataset does not contain key process parameters (such as machine speed, temperature, operator, or material type), limiting root cause analysis to observational trends rather than process-level causality.

2. Literature review

2.1. Defect classification systems

Defect classification systems are an essential component of structured quality control processes, as they enable defects to be systematically identified, categorised, and prioritised according to their severity. In many industrial environments, severity ratings are recorded by human operators, which introduces subjectivity and inconsistency due to differences in experience, judgement, and working conditions. As a result, recent research has increasingly focused on combining human inspection with artificial intelligence (AI) methods to improve defect detection accuracy and classification reliability.

One of the most widely applied approaches is the use of convolutional neural networks (CNNs), which can be trained on defect imagery to automatically classify defect types. Meijer et al. (2019) demonstrated that CNN-based inspection of sewage pipe imagery can significantly reduce classification errors commonly associated with manual inspection. Their study further reported that automation could reduce human labour requirements by approximately 60.5%, resulting in substantial labour cost savings. Similarly, Ye et al. (2018) observed that while automated optical inspection (AOI) systems are effective at detecting the presence of defects, they often struggle to accurately identify detailed defect characteristics such as flaw size and inclination. To address this limitation, their study proposed a CNN-based classification model capable of extracting defect feature details more effectively than conventional AOI systems.

2.2. Cost of poor quality (COPQ)

Prashar (2014) defines the *cost of poor quality (COPQ)* as the total cost associated with failures such as repair, rework, scrap, service calls, warranty claims, and write-offs of obsolete finished goods in manufacturing. COPQ is often used as a performance indicator for quality management, since it reflects the financial impact of defects and process inefficiencies. Mahmood et al. (2014) report that organisations with effective quality control systems may reduce COPQ to approximately 2.5% of sales, whereas higher values, such as 4–5%, may indicate poor process capability. To quantify COPQ, Mahmood et al. (2014) propose the following equation:

$$COPQ = \sum RMT \times MQ + \sum RMH \times MHQ + \sum RMC \times MCQ + \sum RT \times TQ$$

where RMT is the unit rate of material, and MQ is the quantity of lost material; RMH is the labour cost rate per man-hour and MHQ is the quantity of lost man-hours; RMC is the machine-hour rate, and MCQ is the quantity of lost machine hours; and RT represents the overhead cost rate per day while TQ represents the number of days lost on activities along the critical path. This model is useful because it expresses defect-related losses in terms of direct material waste, labour inefficiency, equipment downtime, and schedule disruption.

2.3. Six Sigma DMAIC

As a proven model for refining existing workflows, the DMAIC methodology facilitates significant improvements in organizational performance. According to Prashar (2014), its implementation leads to a measurable reduction in defects and cycle durations. Furthermore, it strengthens customer loyalty and financial health, making it a staple for quality improvement in various global industries. It is said that “six sigma level” companies have a COPQ < 10 percent of sales, about 15 to 20 percent of sales for companies who are at “four sigma” level and about 20 to 30 percent of sales for companies who are at “three sigma” levels. Furthermore, with DMAIC the aim is to stop the defects before they appear and reduce the COPQ by adopting a predictive rather than a reactive approach toward rejection and rework (Prashar, 2014).

2.4. Statistical tools in quality improvement

Statistical tools enable quality practitioners to quantify and monitor process performance. Once quality is quantified, it becomes easier to analyse variation, identify root causes, and implement process improvements. Common statistical tools applied in quality improvement include descriptive statistics, Pareto analysis, trend analysis (run charts), statistical process control (control charts), box plots, hypothesis testing (t-tests, ANOVA and chi-square tests), and regression analysis. Lin et al. (2008) successfully applied the process capability index to evaluate process performance and achieved a reduction in standard deviation through the Taguchi method. Similarly, Kumar et al. (2007) demonstrated the effective use of Pareto analysis, measurement system analysis, regression analysis, and design of experiments within the DMAIC framework. Their findings showed that Six Sigma implementation reduced casting defects and improved the process capability index from 0.49 to 1.28, resulting in a significant financial impact of over US$110,000 per annum.

3. Methodology

3.1. Dataset description

The manufacturing defects dataset used in this study was obtained from Kaggle (author, year). The dataset contains simulated quality control records representing defects detected during manufacturing inspections. The variables capture key defect-related characteristics such as defect classification, detection date, defect location, severity level, inspection method, corrective action, and associated repair costs.

The dataset consists of defect-level observations, where each record represents a unique defect incident linked to a specific product. Although the dataset is not based on a specific real manufacturing company, it provides a suitable structure for applying statistical quality control tools and analysing defect trends, severity distribution, and cost implications.

Variables Included
The dataset contains the following variables: defect_id (unique defect identifier), product_id, defect_type, defect_description, defect_date, defect_location, severity, inspection_method, repair_action, and repair_cost. Repair_cost represents the cost incurred in correcting the defect; however, the dataset documentation does not specify the currency unit.

Application in This Study
This dataset supports statistical analysis aimed at identifying defect patterns, evaluating severity distribution, assessing inspection effectiveness, and estimating the financial impact of defects. The dataset is therefore appropriate for applying tools such as Pareto analysis, trend analysis, descriptive statistics, and regression modelling to identify key defect drivers and recommend quality improvement strategies.

3.2. Data preprocessing steps

Prior to analysis, the dataset was preprocessed using Python to ensure accuracy and consistency. The preprocessing included checking for missing values and duplicate records, validating variable formats, and converting relevant fields such as defect_date into an appropriate datetime format. Numerical variables such as repair_cost were examined for outliers and invalid values, while categorical variables (e.g., defect_type, severity, inspection_method) were checked for inconsistent labels. Where necessary, data cleaning procedures were applied to improve completeness and reliability of the dataset before statistical analysis.

3.3. Tools used

Python with its various Libraries (i.e. Pandas, seaborn,matplotlib, etc.) was applied in Jupyter notebook. Further, a panel dashboard was used.

3.4. Analysis techniques used

Exploratory Data Analysis (EDA) was done as the first step after preprocessing. This ensured there is an understanding of the variables through univariate,bivariate and multivariate analysis. In EDA descriptive statistics such as measure of central tendency and measures of spread were used for variables such as repair_cost. Furthermore, Pareto Analysis, Trend Analysis, Control Chart, and Regression were used.

4. Results and Discussion

4.1. Defect type distribution

The manufacturing defects dataset consisted of three defect categories: structural, functional, and cosmetic defects. Structural defects compromise the physical integrity of the product, rendering it unsuitable for sale. Functional defects occur when the product appears visually acceptable but fails to perform its intended function. Cosmetic defects refer to products that remain functional and structurally sound but fail to meet aesthetic or quality appearance standards.

Figure 1 illustrates the distribution of defect types within the manufacturing process. The results indicate that structural defects occurred more frequently than functional and cosmetic defects. This suggests that significant variability may exist within critical production conditions and process controls.

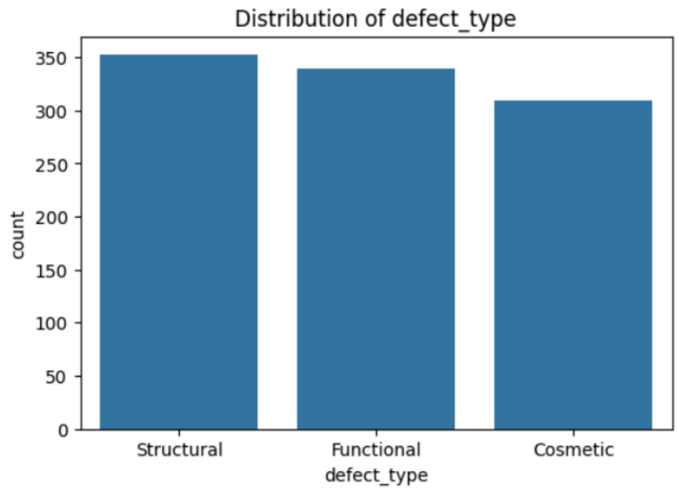

Figure 1: Defect_type distribution

To further evaluate defect contribution, Pareto analysis was conducted using the 80–20 principle, as illustrated in Figure 2. The Pareto chart indicates that structural defects contribute the largest proportion of total defects, followed by functional and cosmetic defects. This finding suggests that prioritising the reduction of structural defects may result in the greatest improvement in overall product quality and process efficiency.

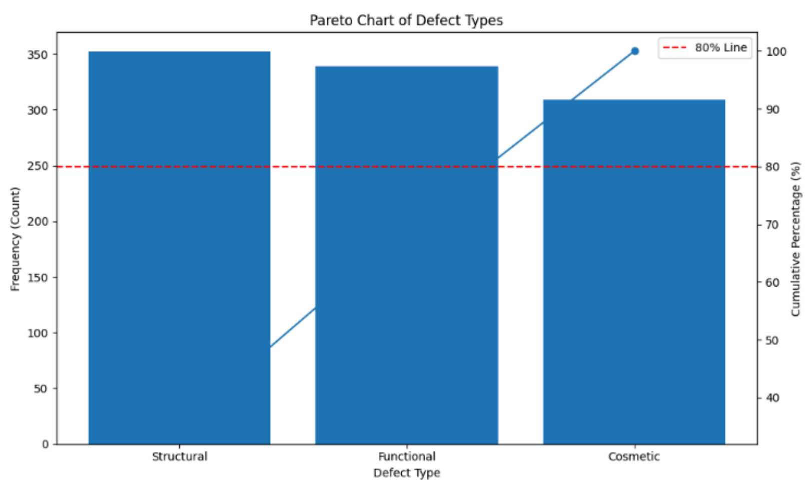

Figure 2: Pareto Chart for defect_type

In addition, control charts were used to evaluate defect variation and process stability over time, as shown in Figure 3. The results indicate that cosmetic defects remained below the process mean of 51 defects for three consecutive months from February, suggesting relatively stable process behaviour during this period. Functional defects similarly remained below the mean for four months, although spikes were observed in March and May 2024, indicating periods of increased process variation.

In contrast, structural defects exhibited a higher process mean and more observations above the mean compared to the other defect categories. This suggests greater instability and variability within the processes associated with structural defects, highlighting the need for further investigation into potential root causes and process control measures.

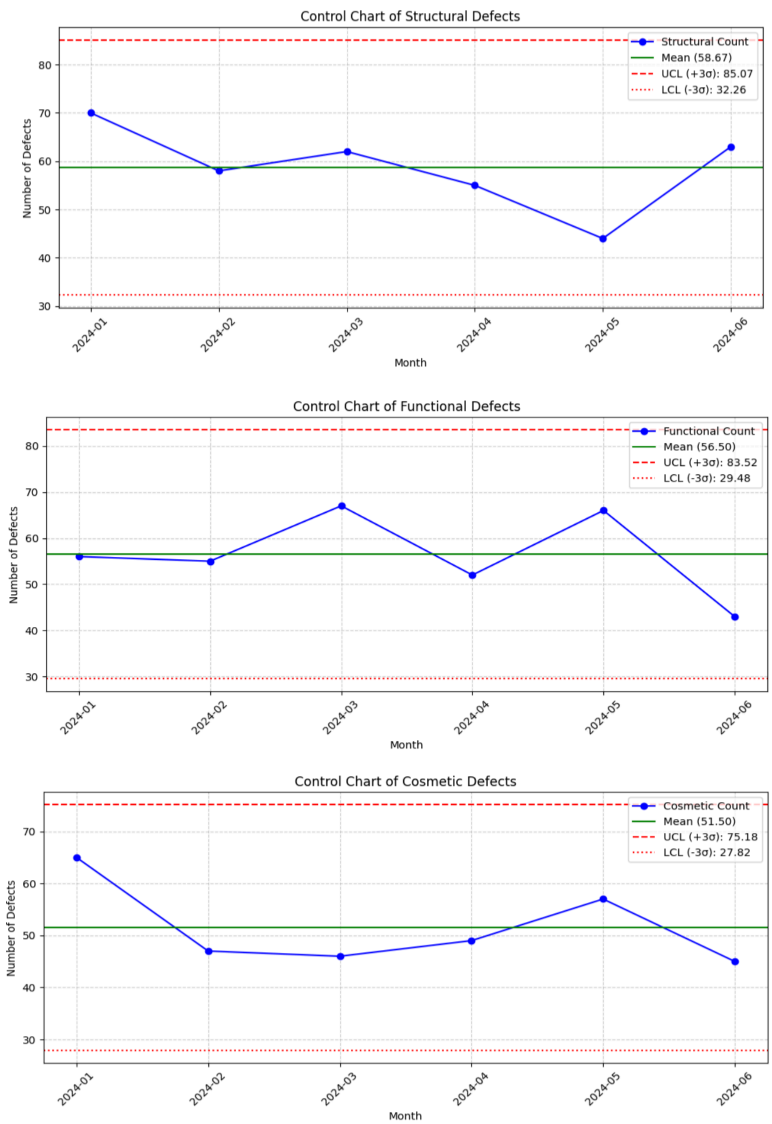

Figure 3: Defect type control charts

\

4.2. Defect trends over time

A trend plot of repair cost over time for the three inspection methods, namely automated, manual, and visual inspection, was generated for the year 2024. The plotted trends exhibited fluctuating patterns across all inspection methods, indicating variability in repair costs over time. No clear relationship between inspection method and repair cost could be conclusively observed from the figure alone. This may be attributed to the fact that inspection methods and repair costs belong to different categories within the Cost of Quality (COQ) framework. Inspection methods are associated with the cost of good quality, while repair costs are associated with the cost of poor quality. Nevertheless, the effectiveness of an inspection method may indirectly influence the quantity of defective products requiring rework or repair.

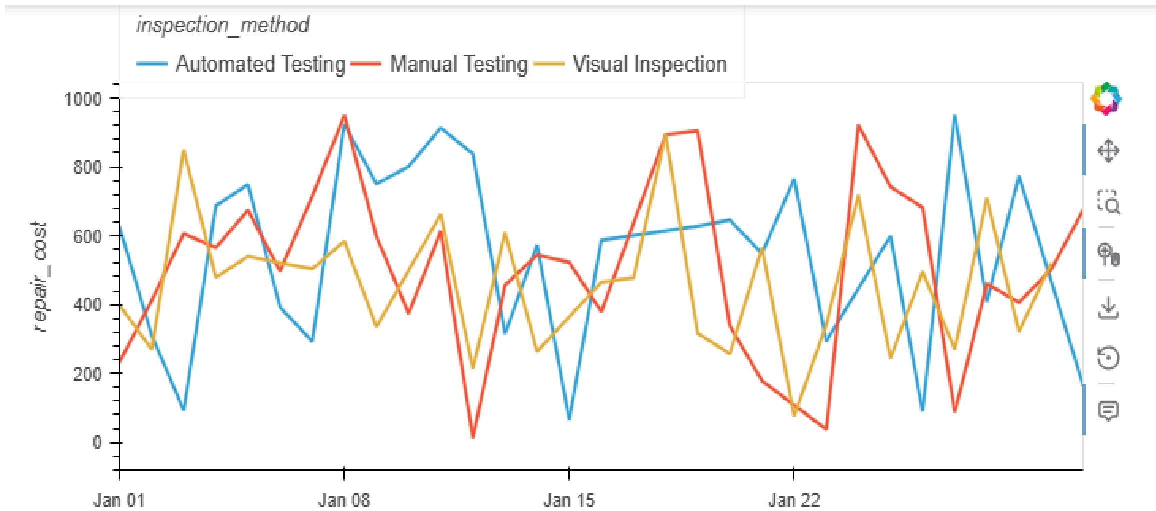

Figure 4: relationship between repair cost and inspection method over time

4.3. Severity distribution

The term "severity" describes how serious, harsh, or damaging a circumstance, occurrence, or condition is. It gauges the severity or importance of a problem, frequently revealing the degree of a detrimental effect or the degree of assistance required.

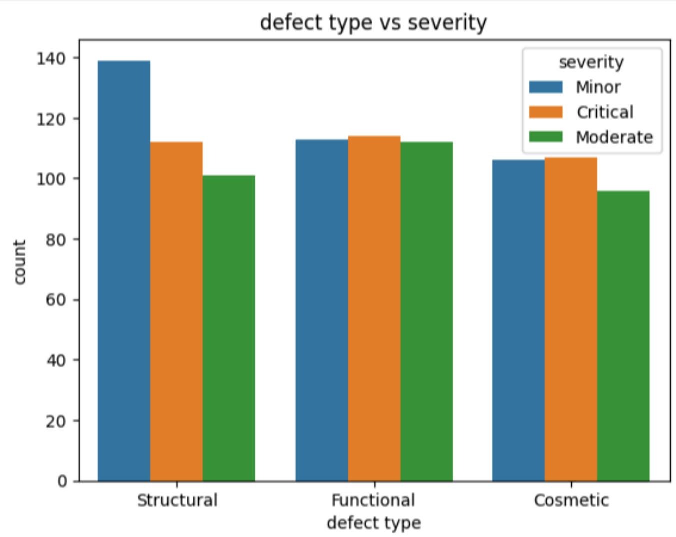

Figure 5: relationship between defect type and severity

Figure 5 illustrates the distribution of severity levels across structural, functional, and cosmetic defects. Structural defects recorded the highest number of minor severity cases, while functional defects showed a relatively balanced distribution across all severity categories. Critical severity defects were also notably high, particularly in structural and functional defect types, suggesting potential safety and reliability concerns within the manufacturing process. Cosmetic defects exhibited comparatively fewer moderate severity cases, indicating that most cosmetic issues were either minor in nature or escalated directly to critical classification. Overall, the results suggest that defect severity varies across defect categories and may require different quality control strategies depending on the nature of the defect.

4.4. Repair cost analysis

Repair cost was analysed against inspection method to determine whether inspection approaches influence cost variability. Although Visual Inspection and Manual Testing exhibited similar data counts, noticeable differences were observed in the repair cost distributions across inspection methods.

The median repair costs were approximately equal for all three inspection methods. However, Visual Inspection displayed a narrower interquartile range, suggesting more consistent repair costs. Manual Testing exhibited greater spread, while Automated Testing showed the widest distribution, indicating higher variability in repair costs.

The lower variability associated with Visual Inspection may be attributed to its limited reliance on specialised equipment. In contrast, Manual and Automated Testing often require additional equipment, operational support, and maintenance resources, which may contribute to wider repair cost variation. The greater variability observed in Automated Testing could also reflect fluctuations associated with machine operation and maintenance requirements.

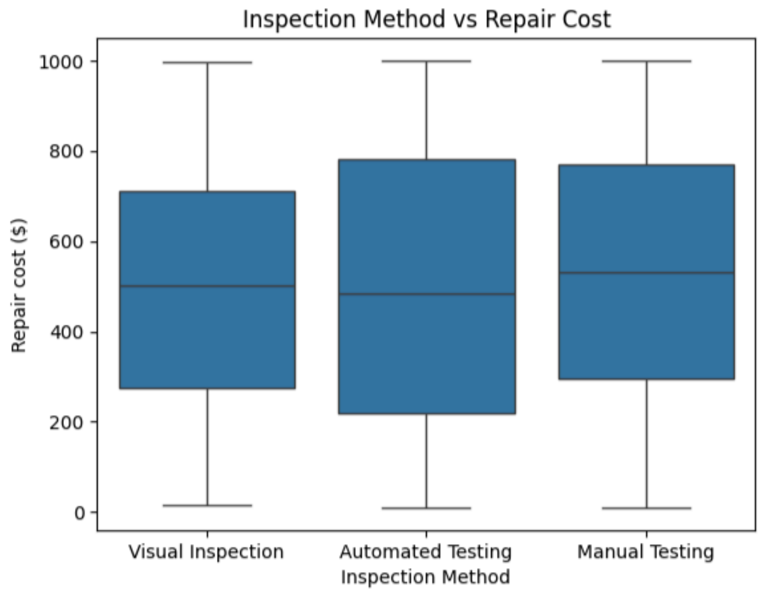

Figure 6: Inspection method versus repair cost

4.5. Inspection method effectiveness

In figure 7 below, the data suggests that Visual Inspection and Manual Testing were the most frequently utilized quality control methods within the manufacturing environment under investigation. This trend may be influenced by several factors, including cost-efficiency, operational flexibility, and existing production infrastructure. Many manufacturing facilities continue to rely on legacy systems and older machinery, which may limit the adoption of capital-intensive automated technologies.

From an economic perspective, the initial investment required for automated inspection systems can be significantly higher than the cost of employing skilled operators. As a result, manufacturers may prefer manual and visual inspection methods, particularly in environments where production costs must be carefully controlled. Furthermore, in high-pressure production settings, experienced technicians are often able to perform rapid visual assessments and manual adjustments with greater flexibility and adaptability.

The higher frequency of Visual Inspection and Manual Testing observed in the dataset may therefore reflect the practical and economic advantages associated with these methods. In contrast, Automated Testing systems often require specialised equipment, technical expertise, and ongoing maintenance support, which may contribute to lower implementation rates despite their potential benefits in process consistency and precision.

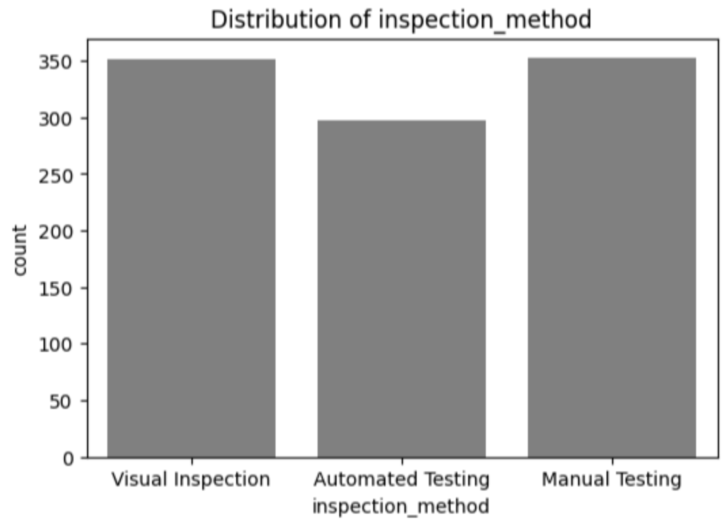

Figure 7: Univariate analysis of inspection methods distribution

<u>4.6. Defect location patterns</u>

Defects are categorized by their location within the product or machinery: Surface, Component, or Internal. Component defects pertain to failures in individual parts or tooling, while internal defects encompass structural flaws incurred during manufacturing or detected within the finished good. Figure 8 illustrates the distribution of these defects across the manufacturing dataset.

Figure 8: Univariate analysis of defect location

The univariate analysis in Figure 8 reveals that surface-level defects are the most prevalent. This prevalence is largely attributable to the ease of detection; surface defects are readily identifiable during standard visual inspection. Conversely, component and internal defects often remain obscured, necessitating specialized testing protocols or, in some instances, being identified only after the product reaches the end user.

To further understand these patterns, the relationship between defect location and defect type was analyzed (Figure 9).

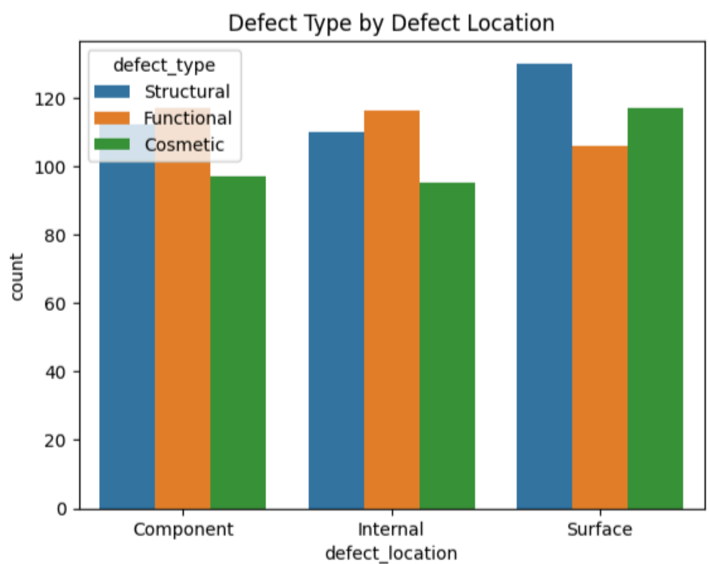

Figure 9: defect location versus defect location.

The data in Figure 9 reveals distinct trends by location:

- **Surface:** Structural defects occur most frequently, followed by cosmetic and functional defects.
- **Component:** Structural and functional defects are more common than cosmetic issues.
- **Internal:** Functional defects are more prevalent than cosmetic defects, likely because internal cosmetic flaws often remain undetected until the product lifecycle progresses.

5. Recommendations

The Six Sigma DMAIC framework is recommended as a suitable methodology for quality control and defect reduction in manufacturing environments. Previous studies have demonstrated its effectiveness in improving process performance and reducing variability. In this study, only the Define, Measure, and Analyze phases of the DMAIC framework were applied due to the generalized nature of the manufacturing dataset used. As a result, the Improve and Control phases could not be fully implemented. Future studies should incorporate real-time industrial data to enable the development, implementation, and monitoring of targeted improvement strategies.

6. Conclusions

The study demonstrates that underutilised defect datasets contain significant intelligence regarding process inefficiencies. By shifting from reactive inspection to data-driven decision-making, manufacturers can prioritize resources toward the most frequent and costly defect drivers. Moving forward, the integration of process-level parameters (such as temperature, machine speed, and operator data) will be essential to transition from identifying observational trends to establishing root-cause causality. Implementing the remaining "Improve" and "Control" phases of the DMAIC cycle, supported by real-time industrial data, remains the logical next step to transform these analytical findings into sustained, long-term defect reduction and optimized operational performance

References

Mahmood, Shahid, et al. “Determining the Cost of Poor Quality and Its Impact on Productivity and Profitability.” *Built Environment Project and Asset Management*, vol. 4, no. 3, 7 July 2014, pp. 296–311, https://doi.org/10.1108/bepam-09-2013-0034. Accessed 27 May 2026.

Meijer, Dirk, et al. “A Defect Classification Methodology for Sewer Image Sets with Convolutional Neural Networks.” *Automation in Construction*, vol. 104, Aug. 2019, pp. 281–298, https://doi.org/10.1016/j.autcon.2019.04.013. Accessed 27 May 2026.

Mukti Ali Sadikin. “Defect Reduction in the Manufacturing Industry: Systematic Literature Review.” *International Journal of Industrial Engineering and Engineering Management*, vol. 5, no. 2, 17 Dec. 2023, pp. 73–83, https://doi.org/10.24002/ijieem.v5i2.7495. Accessed 27 May 2026.

Owen, David. “Issue 4 PRODUCTS LIABILITY LAW SYMPOSIUM in MEMORY of PROFESSOR GARY T. SCHWARTZ Article 8 Summer 2002 Part of the Law Commons Recommended Citation Recommended Citation Owen.” *South Carolina Law Review South Carolina Law Review*, vol. 53, 2002. Accessed 27 May 2026.

Prashar, Anupama. “Adoption of Six Sigma DMAIC to Reduce Cost of Poor Quality.” *International Journal of Productivity and Performance Management*, vol. 63, no. 1, 7 Jan. 2014, pp. 103–126, https://doi.org/10.1108/ijppm-01-2013-0018. Accessed 27 May 2026.

Tiwari, Divya, et al. “In-Process Monitoring in Electrical Machine Manufacturing: A Review of State of the Art and Future Directions.” *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture*, vol. 235, no. 13, 12 May 2021, pp. 2035–2051, https://doi.org/10.1177/09544054211016675. Accessed 27 May 2026.

Ye, Ruifang, et al. “) 1-7 Ó the Author(S.” *Special Issue Article Advances in Mechanical Engineering*, vol. 10, no. 3, 2018, p. 2018, https://doi.org/10.1177/1687814018766682. Accessed 27 May 2026.